

# Zihan Yang

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## About Me

I am a master's student at the School of Aeronautic Science and Engineering, Beihang University, China. My research has covered both learning-based planning and control, and control-theoretic methods to improve learning. Recent works have been accepted at ICLR 2025 (Oral) and ICCA 2024 (Best Student Paper).

## Education

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| <b>Northwestern Polytechnical University</b> , B.Eng. in Material Science and Engineering                               | Sept 2019 - June 2023               |
| <b>Queen Mary University of London</b> B.Eng. in Material Science and Engineering<br>(UK-China Transnational Education) | Sept 2019 - June 2023               |
| • GPA: 3.6/4.0<br>• <b>First Class Honors.</b>                                                                          |                                     |
| <b>Beihang University</b> , M.S. in Aerospace Science and Technology                                                    | Sept 2023 - June 2026<br>(expected) |
| • GPA: 3.8/4.0, supervised by Prof. Kexin Guo, Prof. Xiang Yu, and Prof. Lei Guo                                        |                                     |

## Publications

(\* equal contribution)

- [1] **Z. Yang**, J. Jia, M. Wang, Y. Liu, K. Guo, and X. Yu, "Towards non-structural disturbance prediction: Feedback-calibrated meta-adaptation," *Under Review*, 2025.
- [2] J. Jia\*, M. Wang\*, **Z. Yang**, B. Yang, Y. Liu, K. Guo, and X. Yu, "Learning-based observer for coupled disturbance," 2025.
- [3] J. Jia\*, **Z. Yang\***, M. Wang, K. Guo, J. Yang, X. Yu, and L. Guo, "Feedback favors the generalization of neural ODEs," in *The Thirteenth International Conference on Learning Representations*, **Oral Presentation (acceptance rate: 1.8%)**, 2025.
- [4] K. Guo\*, **Z. Yang\***, J. Jia, Y. Liu, and X. Yu, "Optimizing control-friendly trajectories with self-supervised residual learning," *Under Review*, 2024.
- [5] **Z. Yang**, J. Jia, Y. Liu, K. Guo, X. Yu, and L. Guo, "TRACE: Trajectory refinement with control error enables safe and accurate maneuvers," in *IEEE 18th International Conference on Control and Automation*, **Best Student Paper Award**, 2024.
- [6] X. Song, X. Guo, G. Liu, **Z. Yang**, and L. Liu, "Co-optimization of motion and energy domain for hydrogen-powered hybrid UAVs: A bi-directional coupling architecture," in *IEEE 19th International Conference on Control and Automation*, 2025.
- [7] K. Guo, Y. Liu, J. Jia, **Z. Yang**, S. Zhou, X. Yu, and L. Guo, "Flying in harsh environments: Anti-rain quantification system and control strategy," *IEEE Transactions on Industrial Electronics*, 2025.

## Honors and Awards

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|------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Oral Presentation, Top 1.8%</b> The Thirteenth International Conference on Learning Representations (ICLR), Singapore.    | April, 2025 |
| <b>Best Student Paper Award</b> , <i>IEEE</i> International Conference on Control and Automation (ICCA), Reykjavík, Iceland. | June, 2024  |

## Funds

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|---------------------------------------------------------------------|------|
| <b>Second-Class Scholarship</b> Beihang University, Beijing, China. | 2024 |
| <b>Second-Class Scholarship</b> Beihang University, Beijing, China. | 2023 |

## Skills

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**Coding:** Python (skilled), Matlab/Simulink (skilled), C/C++ (familiar), HTML/CSS (basic)

**Language:** Mandarin Chinese (native), English (fluent), Japanese (basic)

**Hardware:** Experienced in building quadrotor hardware modules

**Misc.:** Electric guitar, skiing, badminton, swimming...

**Soft Skills.:** Self-motivated, critical-thinking, multi-tasking, and a quick learner...